Session 3: Emerging AI Trends in Earth Observation

Foundation Models, Embeddings, Self-Supervised Learning, On-board AI and Explainable AI

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Agenda & Pacing

- **Duration:** 2 hours (120 minutes)
- **Plan:**
  - 0–10: Session goals & context
  - 10–55: Geospatial foundation models (GeoFMs) and embedding datasets
  - 55–70: On-board AI (Φsat-2, models in orbit)
  - 70–90: Self-supervised learning (SSL)
  - 90–110: Explainable AI (XAI)
  - 110–120: Discussion & next steps

> **Note**
>
> State of the field as of **November 2026**. This area moves fast: every model fact on these slides carries a source, so check the link for the latest version.

Learning Objectives

- Define foundation models and name the main open GeoFMs (Prithvi-EO-2.0, Clay, TerraMind, Copernicus-FM, DOFA, SatMAE)
- Explain **embedding datasets** (AlphaEarth Foundations / Satellite Embedding, Tessera, OlmoEarth) and when to use them instead of fine-tuning
- Describe what on-board AI can do today (Φsat-2, GeoFMs in orbit)
- Explain SSL and why it suits unlabelled satellite archives
- Apply XAI concepts (SHAP, LIME, Grad-CAM, Integrated Gradients) to EO models
- Decide when FM, embeddings, SSL or XAI fit a Philippine use case

Foundation Models (GeoFMs)

What & Why

- Pre-trained once with **self-supervision** on large, global EO archives
- Adapted to many tasks: **fine-tune** a head (or the whole model), or just use the **embeddings** as features
- Label-efficient: published benchmarks show useful results from small labelled sets, but how small depends on the task. Test on your own data before promising “100 labels”
- Global pre-training data includes Southeast Asia, but check how well your region and season are represented

Traditional CNN vs Foundation Model

	Train a CNN from scratch	Start from a GeoFM
Pre-training	none (or ImageNet)	self-supervised on global EO archives
Labels needed	many (thousands typical)	fewer: start small, measure, add more
Sensors	usually one, fixed bands	model-dependent: multi-sensor, any-band models exist
Compute to adapt	full training run	fine-tune a head / small model, or train a classifier on embeddings
Main risk	overfitting small data	domain shift (sensor, resolution, region) from the pre-training data

Why It Matters for the Philippines

- **Label scarcity:** few agencies hold large, consistent labelled datasets
- **Disaster tempo:** typhoon and flood products are needed within hours to days
- **Heterogeneous landscapes:** from Metro Manila to BARMM, plus persistent cloud cover (SAR-capable models help)
- **Budget efficiency:** one backbone (or one embedding dataset) reused for crops, mangroves and settlements

Open GeoFMs You Can Use (Nov 2026)

Model	Who / when	What it is	Source
Prithvi-EO-2.0	IBM, NASA, Jülich SC · Dec 2024	300M / 600M ViT, masked autoencoder on **4.2M** global HLS time-series samples (30 m), temporal + location embeddings	arXiv 2412.02732, IBM blog
Clay v1.5	Clay · 2024 (trained Sep 2024)	MAE-based, any number of bands; trained on Sentinel-2, Landsat, Sentinel-1, NAIP, LINZ, MODIS; Apache-2.0	Clay v1.5 spec
TerraMind 1.0	IBM + ESA Φ-lab + Jülich SC · Apr 2025	any-to-any **generative** multimodal model (S1, S2, DEM, land cover…), >9M samples; tiny/small edge versions Oct 2025	ESA, Φ-lab
Copernicus-FM	TUM and partners · ICCV 2025	extends DOFA to any Sentinel modality (S1, S2, S3, S5P); with Copernicus-Pretrain data and Copernicus-Bench	GitHub
DOFA	Xiong et al. · 2024	“Dynamic One-For-All”: wavelength-conditioned patch embedding, so one model serves optical, SAR, hyperspectral	arXiv 2403.15356
SatMAE	Stanford · NeurIPS 2022	early MAE for temporal + multispectral imagery (fMoW-Sentinel); a key ancestor of the models above	project page

Fine-tuning toolkit for Prithvi/TerraMind and many others: TerraTorch

The 2025–26 Shift: Embedding Datasets

Instead of running a model yourself, **download pre-computed embeddings** (one vector per 10 m pixel per year) and train a small classifier on them, like the Random Forest from Day 2.

Dataset	Producer	Details	Source
Satellite Embedding V1 (AlphaEarth Foundations)	Google / Google DeepMind · Jul 2025	64-D, 10 m, annual, global land + coastal; in Earth Engine as `GOOGLE/SATELLITE_EMBEDDING/V1/ANNUAL` (bands `A00` – `A63`)	EE catalog, DeepMind, paper
Tessera	Univ. of Cambridge and partners · 2025 (CVPR 2026 paper)	Sentinel-1 + Sentinel-2, 10 m, annual embeddings 2017–2025, open, no registration	ESA, Jun 2026
OlmoEarth v1.3	Ai2 · Oct 2026	S1 + S2 + Landsat year → 128-D vector per 10 m pixel (first 64 dims usable alone); global embeddings on Source Cooperative	Ai2 blog

Caveat: embedding dimensions are not physically interpretable. Validate against local reference data before operational use.

Embeddings in Practice (Earth Engine)

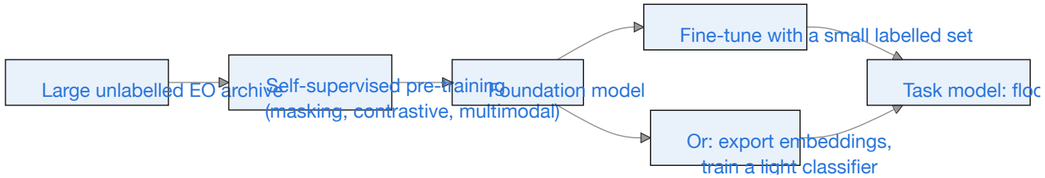
```python
import ee
ee.Initialize(project="your-cloud-project")   # EE now requires a Cloud project

aoi = ee.Geometry.Rectangle([124.36, 7.05, 125.63, 8.84])   # ~Bukidnon (bbox)
emb = (ee.ImageCollection("GOOGLE/SATELLITE_EMBEDDING/V1/ANNUAL")
         .filterDate("2024-01-01", "2025-01-01")
         .filterBounds(aoi)
         .mosaic())                                        # 64 bands: A00 … A63

training = emb.sampleRegions(collection=labelled_points,   # your field labels
                             properties=["class"], scale=10)
clf = ee.Classifier.smileRandomForest(100).train(training, "class", emb.bandNames())
landcover = emb.classify(clf).clip(aoi)
```

Same workflow as Day 2, but the 64 embedding bands replace the hand-made spectral features. Google's tutorial series covers this in more detail.

How a GeoFM Works



- Pre-train once (expensive, done by the model provider); adapt many times (cheap)

Philippine Use Cases: Project Ideas

These are **illustrative project ideas** for discussion, not deployed systems. The numbers depend on your data.

- **Rapid flood mapping:** fine-tune Prithvi-EO-2.0 or TerraMind on Sentinel-1/-2 flood labels (start from public Sen1Floods11 checkpoints, add local Philippine events)
- **Crop and land-cover mapping:** Satellite Embedding / Tessera features + a few hundred DA/PSA field points → provincial crop map
- **Mangrove monitoring:** embeddings + DENR / NAMRIA reference plots, compared year to year for change
- **Drought (link to Session 2):** add embedding-derived land-cover masks to restrict NDVI forecasts to cropland

On-board AI

AI in Orbit: Where We Are (Nov 2026)

- **Φsat-2 (ESA):** 6U CubeSat launched 16 Aug 2024 (Transporter-11); multispectral imager; entered operational phase in July 2025
  - On-board apps: cloud detection (discard cloudy scenes), image-to-street-map for disaster response, deep image compression, vessel detection, wildfire and marine-pollution monitoring
  - Source: ESA EO Gateway, Jul 2025
- **GeoFMs in orbit:** a compressed Prithvi model ran flood and cloud detection on the Kanyini satellite and on Thales Alenia Space's IMAGIN-e payload on the ISS, the first GeoFM deployed in orbit (NASA, May 2026, preprint)
- **Edge-sized models:** tiny/small TerraMind and Prithvi variants for laptops, phones and satellites (Oct 2025, IBM)

Why it matters: less downlink and lower latency, which helps time-critical alerts (fires, floods, ships). Trade-off: smaller models, harder validation.

Self-Supervised Learning

Why SSL for EO

- Only a tiny fraction of the satellite archive is labelled; SSL learns from the raw archive
- Cuts annotation cost and helps “cold-start” regions with few labels
- Produces representations reusable for segmentation, detection, forecasting, which is what makes GeoFMs possible

Common SSL Strategies

- **Masked autoencoding:** hide most patches (often ~75%) and reconstruct them (SatMAE, Prithvi, Clay)
- **Contrastive learning:** pull together views of the same place/time, push apart others (SimCLR, MoCo); **self-distillation** variants need no negatives (BYOL, DINO)
- **Temporal pretext tasks:** order scenes, predict the next time step, detect anomalies
- **Cross-modal learning:** align or generate Sentinel-1 ↔ Sentinel-2 ↔ DEM (TerraMind, Copernicus-FM)

Building an SSL Pipeline

1. Collect unlabelled Sentinel/Landsat imagery for your area of interest
2. Cloud/shadow filtering; tile into patches or time series
3. Choose a pretext objective (masked, contrastive, temporal)
4. Pre-train a backbone (ViT, CNN, LSTM) on GPUs
5. Fine-tune on mission-specific labels

In practice: before pre-training your own model, check whether an existing GeoFM or embedding dataset already covers your sensor and region.

Explainable AI (XAI)

Why XAI

- Builds trust for operational agencies (DRRM, agriculture, enforcement)
- Reveals bias or drift (seasonal, regional, sensor)
- Supports governance and documentation requirements

Techniques at a Glance

Technique	Works With	Insight Provided
SHAP	Tree models, tabular and deep models	Feature contribution per prediction
LIME	Any classifier/regressor	Local surrogate explaining one sample
Grad-CAM	CNN-based models	Heatmap of salient pixels/regions
Integrated Gradients	Differentiable models (incl. transformers, LSTMs)	Attribution along an input path

Libraries: SHAP, LIME, Captum (PyTorch)

Applying XAI in EO

- **Flood mapping:** Grad-CAM over segmentation outputs to check the model looks at water, not shadows
- **Drought LSTM (Session 2):** Integrated Gradients to see which past months drive the forecast
- **Crop yield models:** SHAP to rank NDVI, rainfall, soil moisture drivers
- **Vessel alerts:** LIME on trajectory features to explain why a track was flagged
- Pair XAI outputs with review by local analysts (LGUs, NAMRIA, PAGASA)

Philippine Integration Roadmap

1. **Assess:** audit data readiness, choose priority missions (floods, crops, mangroves)
2. **Adapt:** try embeddings first (cheapest); fine-tune a GeoFM if accuracy is not enough; label locally
3. **Deploy:** package as an API/dashboard; share models through national repositories such as DOST-ASTI's DIMER
4. **Scale:** monitor drift with XAI, refresh labels, expand to new regions

Key Takeaways

- **GeoFMs:** open models (Prithvi-EO-2.0, Clay, TerraMind, Copernicus-FM) cut the labels you need; how much depends on the task
- **Embeddings first:** Satellite Embedding, Tessera or OlmoEarth + a Random Forest is the cheapest way to start
- **On-board AI** is operational (Φsat-2) and GeoFMs have run in orbit: lower latency, harder validation
- **SSL** is what makes all of this possible; check existing models before pre-training your own
- **XAI** builds trust and catches models that look at the wrong thing; pair it with expert review

Discussion & Q&A

- Where can GeoFMs or embeddings help your agency most?
- Data readiness and labelling strategies
- On-board vs ground processing: what latency do you actually need?
- Risks, governance, and explainability

Sources

- Prithvi-EO-2.0: Szwarcman et al., arXiv:2412.02732; IBM Research blog; Hugging Face
- Clay v1.5: model specification
- TerraMind: ESA, 22 Apr 2025; ESA Φ-lab, Nov 2025; Hugging Face
- Copernicus-FM: Wang et al., ICCV 2025, GitHub · DOFA: Xiong et al., arXiv:2403.15356 · SatMAE: Cong et al., NeurIPS 2022
- AlphaEarth Foundations: Brown et al., arXiv:2507.22291; Satellite Embedding V1
- Tessera: ESA, 4 Jun 2026 · OlmoEarth v1.3: Ai2, 22 Oct 2026
- Φsat-2: ESA EO Gateway · Prithvi in orbit: NASA, May 2026